1(a) 组(1):

比较 $f_1(n)$ $f_2(n)$

$$\frac{f_1(n)}{n} = \frac{\log_2 n}{n^{0.00001}} \xrightarrow{n\to\infty} 0$$

$$f_1(n) < f_2(n)$$

$$f_4(n) = \Theta(n^2)$$

$f_3(n)$ 为指数增长

故 $f_1(n) < f_2(n) < f_4(n) < f_3(n)$

1(b) 组(2)

$$f(1)n = \Theta(1)$$

$$f(4)n = n^{\frac{4}{3}} = \Theta(n^{\frac{4}{3}})$$

$$f_3(n) = \frac{n(n-1)(n-2)}{6} = \Theta(n^3)$$

$$f_2(n) = (2^{20000})^n$$

$\therefore$ $f_1(n) < f_4(n) < f_3(n) < f_2(n)$

1(c) 组(3) $f_4(n) = 2\sum_{i=1}^{n} i + 3n = n^2 + 4n = \Theta(n^2)$.

$f_1(n) = n^{2\sqrt{n}}$ 比固定次数多项式要大 但慢于真正指数函数

$$f_4(n) < f_1(n)$$

再比较 $f_1(n)$ $f_2(n)$ $f_3(n)$ 同时取对

$2\sqrt{n}\ln n$ $\quad n\ln 3$ $\quad 15\ln n + \frac{\ln 2}{3} n$

$\therefore$ $f_1(n) < f_3(n) < f_2(n)$

$\therefore$ $f_4(n) < f_1(n) < f_3(n) < f_2(n)$

2(a): $T(x,y) = \theta(x+y) + \theta(\frac{x}{3}+\frac{y}{3}) + \theta(\frac{x}{9}+\frac{y}{9}) \cdots$

$= \theta(x+y)(1+\frac{1}{3}+\frac{1}{9}+\cdots)$

等比数列相加边界$\frac{3}{2}$.

$\therefore T(x,y) = \theta(x+y)$

2(b): $T(x,y) = \theta(y) + \theta(\frac{y}{4}) + \cdots T(x,c)$

$= \theta(y)(1+\frac{1}{4}+\frac{1}{16}+\cdots) + \theta(x)$

等比数列相加边界$\frac{4}{3}$.

$T(x,y) = \theta(x+y)$

2(c) $T(x,y) = \theta(y) + S(x, \frac{y}{2})$.

$S(x, \frac{y}{2}) = \theta(x) + 2T(\frac{x}{2}, \frac{y}{2})$

$T(x,y) = \theta(x+y) + 2T(\frac{x}{2}, \frac{y}{2})$.

深度为: $k = \theta(\log(\min(x,y)))$

第i层代价: $2^i \theta(\frac{x+y}{2^i}) = \theta(x+y)$

故每层都是: $\theta(x+y)$

递归树总有$\theta(\log\min(x,y))$

$T(x,y) = \theta((x+y)\log(\min(x,y)))$